Replication Debate Teacher Notes

TOK Cognitive Science Activity Bank • Human Sciences • 35-45 min

Category	Human Sciences	**Time**	35-45 min
Difficulty	Extended	**ID**	replication-debate

Big idea: Human science knowledge is context-sensitive.

Overview

Students evaluate what it means for a human-science claim to be supported.

Student Prompt

Inspect Replication Debate Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Replication Debate Stimulus A	Student-facing stimulus 1: A headline claim, a replication attempt, and a contested interpretation. Predict how people will respond, then identify the wording, context, incentive, or role that could change the result.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Replication Debate Stimulus B	Student-facing stimulus 2: Role cards for original researcher, replicator, journalist, and funding body. Predict how people will respond, then identify the wording, context, incentive, or role that could change the result.	Use this for comparison, a second round, or a partner challenge.
Replication Debate Reveal / Twist	Student-facing stimulus 3: A reliability scale separating failure, boundary condition, and fraud. Predict how people will respond, then identify the wording, context, incentive, or role that could change the result.	Teacher reveal: connect the changed response to the big idea — Human science knowledge is context-sensitive.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Replication Debate Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Replication Debate Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Does failed replication destroy knowledge?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how human science knowledge is context-sensitive.

• Strong response: separates observation from inference.

• Strong response: names a method, source, model, language choice, context, or perspective.

• Strong response: revises confidence when new evidence or a new criterion appears.

• Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

• Prepare a starter stimulus using: A headline claim, a replication attempt, and a contested interpretation.

• Print or project the student response grid before the activity begins.

• Plan a private first-response moment before any group discussion.

Ready-to-Use Examples

• A headline claim, a replication attempt, and a contested interpretation.

• Role cards for original researcher, replicator, journalist, and funding body.

• A reliability scale separating failure, boundary condition, and fraud.

Suggested Timing

Stage	Teacher move	Watch for
1	Give students a short description of the claim.	Assumptions, confidence shifts, evidence claims, and language choices.
2	Groups design a replication study.	Assumptions, confidence shifts, evidence claims, and language choices.
3	They must state sample, procedure, outcome measure, possible confounds and what would count as a failure.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Debate whether one failed replication should change belief.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

• Write your first judgement before we discuss it; the first answer is useful evidence.

• Separate what you noticed from what you inferred.

• What would make this knowledge claim more reliable?

Common Misconceptions

• Students may treat class behaviour as universal human nature.

• Students may ignore ethical limits or context effects in human-subject inquiry.

Assessment Look-Fors

• Connects behaviour to method, context, incentives, and ethical constraints.

• Avoids overgeneralising from one small sample.

Differentiation and Adaptations

- Short lesson: use only the first example, one response grid, and one debrief question.
- Long lesson: add a second round where students revise the method and compare outcomes.
- Low-tech option: run with printed cards, sticky notes, and a board tally.

Extension Tasks

- Design a second version of the activity that tests the same knowledge issue in another AOK.
- Find a real-world example where the same knowledge problem matters outside the classroom.
- Write a 150-word TOK exhibition-style commentary using the activity as the object context.

Related Activities

- Ultimatum Game
- Survey Wording Experiment
- Conformity without Humiliation